

Foundations of Image to image translation

[Image to image translation]

1.0 Content Block

Motion and interactive segmentation Motion based segmentation is a technique that relies on motion in the image to perform segmentation. The idea is simple: look at the differences between a pair of images. Assuming the object of interest is moving, the difference will be exactly that object. Improving on this idea, Kenney et al. proposed interactive segmentation [2]. They use a robot to poke objects in order to generate the motion signal necessary for motion-based segmentation. Interactive segmentation follows the interactive perception framework proposed by Dov Katz [3] and Oliver Brock [4]. Another technique that is based on motion is rigid motion segmentation.

2.0 Content Block

Application of Isolated Point Detection in X-ray Image Processing The detection of isolated points has significant applications in various fields, including X-ray image processing. For instance, an original X-ray image of a turbine blade can be examined pixel-by-pixel to detect porosity in the upper-right quadrant of the blade. The result of applying an edge detector's response to this X-ray image can be approximated. This demonstrates the segmentation of isolated points in an image with the aid of single-pixel probes.

3.0 Content Block

$\operatorname{argmin}_{u,K} \gamma |K| + \mu \int_{\Omega} |K^C| |\nabla u|^2 dx + \int_{\Omega} (u-f)^2 dx.$ $\{\displaystyle \operatorname{argmin}_{u,K} \gamma |K| + \mu \int_{\Omega} |K^C| |\nabla u|^2 dx + \int_{\Omega} (u-f)^2 dx.\}$

4.0 Content Block

Model-based segmentation The central assumption of model-based approaches is that the structures of interest have a tendency towards a particular shape. Therefore, one can seek a probabilistic model that characterizes the shape and its variation. When segmenting an image, constraints can be imposed using this model as a prior. Such a task may involve (i) registration of the training examples to a common pose, (ii) probabilistic representation of the variation of the registered samples, and (iii) statistical inference between the model and the image. Other important methods in the literature for model-based segmentation include active shape models and active appearance models.

5.0 Content Block

$\frac{\partial^2 f(x,y)}{\partial x^2} = f(x+1,y) + f(x-1,y) - 2f(x,y)$ $\{\displaystyle \frac{\partial^2 f(x,y)}{\partial x^2} = f(x+1,y) + f(x-1,y) - 2f(x,y)\}$

6.0 Content Block

Groups of image segmentation Semantic segmentation is an approach detecting, for every pixel, the belonging class. For example, in a figure with many people, all the pixels belonging to persons will have the same class id and the pixels in the background will be classified as background. Instance segmentation is an approach that identifies, for every pixel, the specific belonging instance of the object. It detects each distinct object of interest in the image. For example, when each person in a figure is segmented as an individual object. Panoptic segmentation combines both semantic and instance segmentation. Like semantic segmentation, panoptic segmentation is an approach that identifies, for every pixel, the belonging class. Moreover, like in instance segmentation, panoptic segmentation distinguishes different instances of the same class.

7.0 Content Block

Content-based image retrieval Machine vision Medical imaging, and imaging studies in biomedical research, including volume rendered images from computed tomography, magnetic resonance imaging, as well as volume electron microscopy techniques such as FIB-SEM. Locate tumors and other pathologies Measure tissue volumes Diagnosis, study of anatomical structure Surgery planning Virtual surgery simulation Intra-surgery navigation Radiotherapy Digital Pathology and Histopathology. Nuclei instance segmentation in variously stained whole slide images (WSIs) refers to the automatic delineation of individual cells' nuclei borders. This task is a specialized subproblem of instance segmentation, with high practical importance in biomedical research and diagnostics (e.g. cell counting, morphometric feature extraction, tumor grading, prognostic biomarker extraction). The modern approach to the problem is the use of deep learning architectures developed to encompass the main challenges of separating overlapping or fused nuclei. The availability of few humanly-annotated datasets poses another challenge to the problem. Object detection Pedestrian detection Face detection Brake light detection Locate objects in satellite images (roads, forests, crops, etc.) Recognition Tasks Face recognition Fingerprint recognition Iris recognition Prohibited Item at Airport security checkpoints Traffic control systems Video surveillance Video object co-segmentation and action localization Several general-purpose algorithms and techniques have been developed for image segmentation. To be useful, these techniques must typically be combined with a domain's specific knowledge in order to effectively solve the domain's segmentation problems.

8.0 Content Block

where α is the penalty for change in pixel label and β is the penalty for difference in label between neighboring pixels and chosen pixel. Here $N(i)$ is neighborhood of pixel i and δ is the Kronecker delta function. A major issue with ICM is that, similar to gradient descent, it has a tendency to rest over local maxima and thus not obtain a globally optimal labeling scheme.

9.0 Content Block

The simplest method of image segmentation is called the thresholding method. This method is based on a clip-level (or a threshold value) to turn a gray-scale image into a binary image. The key of this method is to select the threshold value (or values when multiple-levels are selected). Several popular methods are used in industry including the maximum entropy method, balanced histogram thresholding, Otsu's method (maximum variance), and k-means clustering. Recently, methods have been developed for thresholding computed tomography (CT) images. The key idea is that, unlike Otsu's method, the thresholds are derived from the radiographs instead of the (reconstructed) image. New methods suggest the use of multi-dimensional, fuzzy rule-based, non-linear thresholds. In these approaches, the decision regarding each pixel's membership in a segment is based on multi-dimensional rules derived from fuzzy logic and evolutionary algorithms, considering factors such as image lighting, environment, and application.

10.0 Content Block

Dual clustering method This method is a combination of three characteristics of the image: partition of the image based on histogram analysis is checked by high compactness of the clusters (objects), and high gradients of their borders. For that purpose two spaces have to be introduced: one space is the one-dimensional histogram of brightness $H = H(B)$; the second space is the dual 3-dimensional space of the original image itself $B = B(x, y)$. The first space allows to measure how compactly the brightness of the image is distributed by calculating a minimal clustering k_{min} . Threshold brightness T corresponding to k_{min} defines the binary (black-and-white) image – bitmap $b = \phi(x, y)$, where $\phi(x, y) = 0$, if $B(x, y) < T$, and $\phi(x, y) = 1$, if $B(x, y) \geq T$. The bitmap b is an object in dual space. On that bitmap a measure has to be defined reflecting how compact distributed black (or white) pixels are. So, the goal is to find objects with good borders. For all T the measure $MDC = G/(k \times L)$ has to be calculated (where k is difference in brightness between the object and the background, L is length of all borders, and G is mean gradient on the borders). Maximum of MDC defines the segmentation.